

Predicting Acoustic Absorption In Additively Manufactured Porous Microlattices: A Sensitivity Analysis Approach

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The study explores the use of sensitivity analysis based on surrogate modeling to predict the absorption coefficient of porous microlattices fabricated through additive manufacturing. Unlike conventional analytical or numerical models, the proposed approach considers the impact of additive manufacturing imprecision on acoustic parameters, which is often overlooked when discrepancies between models and experimental results arise. Additive manufacturing, while offering unparalleled design flexibility, introduces non-trivial defects in the microlattice structures. The manufacturing process typically leads to three main types of defects: elliptical filament sections, filament section shrinkage, and surface roughness. These imperfections can significantly influence the acoustic properties of the porous material, leading to deviations from the idealized geometry assumed in conventional modeling approaches. In this work, sensitivity analysis using Morris and Sobol indices is employed to identify the most influential manufacturing defects affecting the absorption coefficient of the porous microlattice. To address the challenges posed by additive manufacturing imprecision, a metamodel is developed, which efficiently combines the experimental data obtained from fabricated samples with numerical simulations accounting for various defect scenarios. The metamodel serves as a powerful tool for cost-effective and accurate prediction of acoustic properties, even for untested configurations. The sensitivity analysis identifies critical manufacturing parameters and offers valuable insights into the dominant sources of variability. The approach offers a comprehensive framework for quantifying the impact of each additive manufacturing defect on the acoustic performance of porous microlattices, thus prioritizing improvement efforts in the manufacturing process. The scientific outcomes contribute to understanding how additive manufacturing influences the acoustic behavior of porous materials, facilitating the development of more accurate predictive models and informed design choices. The proposed methodology can have significant implications in diverse applications, including acoustic insulation, sound absorption, and noise control in various engineering and architectural fields.

On The Use Of Weakly Coupled Absorber For Low Frequency Sound Absorption

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Weakly coupled systems have promising potential in the design of low-frequency sound absorption systems. Traditional sound absorption techniques often struggle to effectively address both low-frequency sounds and a large frequency band. However, weakly coupled systems can utilize a combination of various sub wavelength resonant absorber to mitigate this challenge. By incorporating elements such as multiple layers and air gaps, these systems can be coupled to achieve enhanced absorption of low-frequency sounds. One key advantage of weakly coupled systems is their ability to exploit the phenomenon of mode veering. By strategically designing the dimensions and materials of the subsystem, multiple resonance can be induced at the target low-frequency range. Due to non-symmetry between the modes, a condition of non-superposition of modes is created. At optimal geometrical and material parameters the system have multiple absorption peaks at near identical frequencies creating an absorption plateau. Furthermore, weakly coupled systems offer flexibility in customization. Designers can fine-tune the system's parameters to match specific low-frequency requirements, making them suitable for various applications such as architectural acoustics, automotive noise control, and industrial noise reduction. Additionally, advancements in computational modeling and optimization techniques enable the optimization of weakly coupled systems for improved absorption performance. As research and development in this field continue to progress, weakly coupled systems hold great promise for addressing the challenging task of low-frequency sound absorption and creating quieter environments in a wide range of settings.

Mass-Spring Model For A Resonant Metamaterial At High Sound Pressure Level

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Metamaterial composed of array of resonators, called here resonant metamaterial, are efficient and compact solutions to achieve high sound absorption at low frequencies. The studied resonant metamaterial is composed of a compact linear periodic array of thin air cavities spaced by small air pores. Recently, a mass spring model was developed to study this acoustic metamaterial. The pores are modelled by a mass and the thin cavities by springs. Thermoviscous losses in the metamaterial are considered following an effective fluid approach. This model has

shown good agreement with impedance tube measurements in linear regime (sound pressure levels up to 100dB). For concrete applications (such as aircraft engines), the studied metamaterial could be used under high levels of acoustic excitation. For sound pressure level greater than 100dB acoustic nonlinear effects appear at orifices. Acoustic resistance of orifice increases with the amplitude of acoustic velocity. At high sound pressure levels, the orifice fluid regime changes and vortex forms at both sides of the orifice, leading to an increase of losses. The air flow resistivity of the pores becomes a linear function of the amplitude of the acoustic velocity (Forchheimer's law). The air flow resistivity coefficients are determined by curve fitting over the entire geometry, based on experimental airflow results and computational fluid dynamics calculations at different flow rates. The different approaches show good agreements with each other. The airflow resistivity function is then included in the mass-spring model. This adapted model allows to predict normal incidence sound absorption coefficient at high sound pressure level. The results of the model are compared with experimental measurements made in an impedance tube at different sound pressure levels.

Waste Corn Husk Fibers For Sound Absorption Applications

Umberto Bernardi

In the last decade, noise pollution and global warming and their effects on human health and the environment have received much attention. Building sectors are one of the most important areas for potential improvements. To this end, sound-absorbing and thermal insulation construction materials are being used effectively. Recently, a great deal of interest has arisen in using various natural fibers as sound-absorbing and/or thermal insulation materials. In line with these studies, this work investigates the acoustic absorption and thermal insulation characteristics of corn husk fiber (CHF). The results showed that the samples enjoy excellent noise reduction coefficients of 0.36-60. The Dunn and Davern (DD) model and a modified model of DD based on the Nelder-Mead simplex algorithm were also used to predict the acoustic behavior of the samples. It was found that the proposed model provides very excellent prediction accuracy.

Novel Acoustic Materials Made From Wood Processing Residues

Suzhou Yin

Wood processing residues, such as shaving, chips, sawdust, wool, bark, present interesting sound absorption properties et have a great potential in the manufacturing of bio-based acoustic materials. Panels were made from shaving, wood wool and grinded bark, with 10% polyvinyl acetate resin as adhesive. The dimensions of the panels were 700 x 700 x 25 or 50 mm and their density varied from 150 to 225 kg/m3. The sound absorption coefficients at 1/3 octave frequencies were measured using an acoustic cabin (Mecanum) between 250 and 10000Hz. Based on the coefficients at four or twelve specific frequencies, the NRC (noise reduction coefficient) and SAA (sound absorption average) were estimated according to ASTM C423. It is shown that the density has an important effect on the NRC and the SAA. In general, panels with a density of 175 to 200 kg/m3 present the best sound absorption performance and good mechanical properties for handling. For panels made with shaving and having a thickness of 25 mm, the NRC varies between 0.70-0.80 and the SAA between 0.60-0.75. By doubling the thickness to 50 mm, the NRC and the SAA increased to 0.95 and 0.93, respectively. The panels made with wool and bark of cedar demonstrate similar or even better sound absorption properties. The manufactured panels were also used as core material in a sandwich system composed of a gypsum board and an oriented strand board (wood) as envelope to improve its sound insulation properties. The STC (sound transmission class) of such a system, for a total thickness of 100 mm, are equal or higher than 50 (minimum requirement for walls and floors between dwellings in the building code). The results of this study suggests that the novel panels are a promising alternative to traditional non-renewable acoustic materials.